

THE UNIVERSITY OF OKLAHOMA
GRADUATE COLLEGE

MONOCULAR CAMERA-BASED AUTONOMOUS
SIDEWALK NAVIGATION USING EFFICIENT
MACHINE-LEARNING-DRIVEN IMAGE SEGMENTATION
AND IMAGE-SPACE PATH PLANNING

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MONOCULAR CAMERA-BASED AUTONOMOUS
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LIST OF ABBREVIATIONS

ARM	Advanced RISC Machine (processor architecture)
BEV	Bird's-Eye View
COCO	Common Objects in Context (dataset)
CPU	Central Processing Unit
DT	Distance Transform
DWA	Dynamic Window Approach
EDT	Euclidean Distance Transform
EMA	Exponential Moving Average
FPS	Frames Per Second
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GPU	Graphics Processing Unit
GT	Ground Truth
HUD	Heads-Up Display
IoU	Intersection over Union
IPM	Inverse Perspective Mapping
NMEA	National Marine Electronics Association
ONNX	Open Neural Network Exchange
RGB	Red, Green, Blue (color space)
ROS	Robot Operating System

ABSTRACT

Autonomous sidewalk navigation for micro-mobility platforms benefits from affordable, lightweight hardware. This thesis targets a deployment scenario in which the entire perception and planning stack runs on a CPU-only single-board computer with a single commodity monocular camera—no GPU, no LiDAR, and no multi-camera rig. This thesis presents a modular vision pipeline designed for this constraint, evolved through four design iterations—from BEV skeleton-graph construction to template-approval planning—and provides a systematic comparison of ten path-planning methods spanning four algorithm families (geometric, cost-field, topological, and template/sampling) across bird’s-eye view (BEV) and image-space domains. A semi-supervised teacher–student framework trains a compact SegFormer-B0 student (3.7M parameters) from OneFormer Swin-L pseudo-labels, subsequently fine-tuned on human-corrected masks to achieve IoU 0.964 at 14.0 ms, with no overfitting observed (held-out IoU 0.974 exceeds training-set IoU 0.963).

On 448 hand-annotated campus sidewalk frames across eight video sequences, image-space midpoint planning achieves 14.5 px lateral center error at 3.0 ms per frame—the lowest error and fastest runtime among all ten methods, with the ranking confirmed as statistically significant (Friedman test, $p < 10^{-61}$; details in Chapter 4). However, the midpoint planner is intent-agnostic: it follows the visible road center but cannot execute commanded turns. BEV-domain planning is therefore retained for metric-scale turn compliance, where a GPS-conditioned waypoint-turn planner with a containment safety guard verifies committed paths against the BEV corridor. The bottleneck in BEV processing lies not in the projection itself but in path *discovery* algorithms—skeleton extraction and Dijkstra search consume 734 ms per frame—motivating the shift to efficient template-approval verification (3 —5 ms).

A template-approval planner replaces the noise-sensitive skeleton-graph baseline, reducing mean heading error by 40.6% and halving path-source switches through fixed-arc

verification with family-reuse hysteresis. Temporal stability analysis reveals that image-space midpoint paths exhibit moderate heading jitter (4.5 deg/frame, $p_{95} = 18.5$ deg), motivating the temporal path smoother that is integrated into the deployed pipeline.

A 1,800-frame accepted run sustains 100% template success at a mean segmentation IoU of 0.954 and 9.97 FPS. The full dual-domain architecture—image-space midpoint for straight-ahead following, BEV corridor verification for turns, and temporal smoothing for deployment stability—sustains 9.97 FPS, meeting the 10 Hz real-time requirement for pedestrian-speed platforms. All evaluation is conducted offline on daytime campus sidewalk video from a single university environment; generalization to other settings, lighting conditions, and closed-loop deployment remains future work.

CHAPTER 1

INTRODUCTION

1.1 Motivation

Consider a battery-powered delivery scooter navigating a university sidewalk. The path ahead is cracked asphalt bordered by grass on one side and a brick retaining wall on the other; there are no lane markings, no curb paint, and no high-definition map of the route. The vehicle’s only forward-looking sensor is a commodity RGB camera, and its only computer is a single-board ARM processor drawing under fifteen watts. In this setting, a natural question arises: how should such a platform perceive the traversable corridor ahead and plan a safe trajectory through it?

The economic and social potential of autonomous sidewalk vehicles suggests that this question is worth answering carefully. Last-mile delivery consumes a disproportionate share of logistics costs, and autonomous delivery robots may help reduce this burden while lowering carbon emissions from conventional delivery vans [1]. Assistive mobility scooters could extend independence for individuals with mobility limitations. Campus shuttle vehicles could improve pedestrian flow in dense environments. These applications share a common technical requirement: the ability to perceive the traversable path ahead and generate safe navigation commands in real time, using sensors and compute that fit within the form factor and power budget of a small electric vehicle. Among available sensor modalities, monocular RGB cameras stand out for their low cost (under \$50), low power consumption (under 15 W for the full system), and compact form factor—making them a natural fit for the platforms described above.

Existing approaches to vision-based navigation face limitations in this setting. End-to-end learning policies can map raw images directly to steering commands, but they

typically require GPU-class hardware and offer limited interpretability [2]. Learned bird’s-eye view (BEV) reconstruction methods such as Lift-Splat-Shoot [3] and BEVFormer [4] achieve impressive results in structured driving scenarios, but they generally assume multi-camera rigs and computational budgets that exceed what a single-board computer can provide [5]. Even classical BEV approaches that project a monocular segmentation mask onto a ground plane via homography face a subtler challenge: while the BEV projection itself produces usable masks with adequate coverage, the path *discovery* algorithms that operate on BEV grids—skeleton extraction, Dijkstra search, distance-transform ridge tracing—are computationally expensive, consuming hundreds of milliseconds per frame. This cost motivates a reexamination of where and how BEV-domain reasoning should be employed in a monocular sidewalk navigation system.

1.2 Problem Statement

This thesis investigates autonomous sidewalk-following navigation for micro-mobility platforms equipped with a single forward-facing monocular camera. The core task is to map each video frame to a centered, smooth navigation path through the traversable region, producing control-ready waypoints suitable for real-time execution on embedded hardware. Within this broad objective, the thesis focuses on two open questions. First, for straight-ahead path following, which planning domain yields better accuracy and computational efficiency—bird’s-eye view or image-space? Second, when BEV-domain reasoning is needed for tasks such as corridor validation and turn planning, how should BEV processing be structured to retain its metric-scale benefits while meeting real-time constraints?

The scope of this work is bounded by the following constraints. All evaluation is conducted offline on pre-recorded campus sidewalk video, isolating the perception and planning stages from closed-loop vehicle dynamics. Runtime measurements are performed under CPU-only inference to reflect the computational budget of embedded deployment. The

evaluation corpus consists of six video sequences totaling over 22 000 frames, supplemented by 448 hand-annotated frames with ground-truth masks for quantitative planner and segmentation comparison. No closed-loop vehicle control is exercised during evaluation, ensuring that the reported metrics characterize the perception and planning pipeline in isolation.

1.3 Approach Overview

This thesis develops a modular pipeline that maps monocular camera frames to controller-ready waypoints through interpretable stages: (i) semantic segmentation via a lightweight SegFormer-B0 model [6] trained with a teacher–student framework using OneFormer Swin-L [7] as the teacher, (ii) optional bird’s-eye view (BEV) projection via a fixed calibrated homography, (iii) mask refinement and connected-component filtering, (iv) path extraction via one of ten planning methods spanning BEV and image-space domains, and (v) temporal smoothing and control output.

The system evolved through four design iterations, each backward-compatible with the previous, tracing an arc from expensive path *discovery* to efficient path *verification*:

- **Iteration 1:** BEV skeleton-graph planner with SegFormer-B2 teacher—constructing navigation graphs from noisy BEV pixels.
- **Iteration 2:** BEV distance-transform corridor planner with enhanced mask morphology—replacing graph construction with ridge-based path extraction.
- **Iteration 3:** Image-space midpoint and distance-transform planners with OneFormer Swin-L teacher—eliminating BEV dependency for straight-ahead path following.
- **Iteration 4:** Template-approval planner with GPS-conditioned waypoint-turn planner and containment safety guard—scoring pre-computed arc templates against BEV corridors rather than building paths from scratch.

This work complements several active research threads while addressing gaps that each leaves open. End-to-end steering policies can achieve strong performance but are diffi-

cult to interpret and typically assume GPU-class hardware [2, 8]. Dense monocular BEV reconstruction methods push accuracy but remain challenging to deploy on low-power platforms [5]. Classic semantic-mask corridor following is efficient in structured environments such as agricultural rows but does not explicitly expose intersection topology [9, 10]. The approach taken in this thesis is monocular, transparent, and computationally light enough for CPU-only inference. To our knowledge, no prior work provides a systematic comparison of BEV-domain and image-space path planning for monocular sidewalk navigation, which is the primary empirical contribution of this thesis.

1.4 Contributions

This thesis makes the following contributions:

1. A systematic comparison of ten path-planning methods spanning four algorithm families on 448 hand-annotated frames across eight video sequences, providing statistically significant evidence (Friedman $p < 10^{-61}$) that image-space midpoint planning achieves 14.5 px lateral error at 3.0 ms per frame for straight-ahead following. The evaluation further demonstrates that midpoint planning is intent-agnostic—it cannot execute commanded turns—motivating the dual-domain architecture that retains BEV for turn compliance (Chapter 4).
2. A template-approval planning architecture that replaces costly skeleton-graph construction with efficient verification of pre-computed arc templates against distance-transform corridors, achieving 69.5 ms mean pipeline time (Table 4.13) while eliminating slow per-frame graph construction. The BEV domain is retained for corridor extraction and turn validation rather than being discarded (Chapter 3).
3. A semi-supervised segmentation training recipe using a high-capacity OneFormer Swin-L teacher [7] to generate pseudo-labels for a compact SegFormer-B0 student [6], achieving

IoU of 0.934 at 14.6 ms on CPU, further improved to IoU of 0.964 through fine-tuning on 228 human-corrected masks. The fine-tuned model generalizes well: held-out video IoU (0.974) exceeds training-video IoU (0.963), confirming no overfitting (Chapter 4).

4. A multi-scale offline evaluation protocol that validates findings at four complementary scales—448 hand-annotated frames across eight video sequences for planner accuracy and segmentation quality, intent-conditioned evaluation on 17 930 annotated video frames for turn compliance, temporal stability analysis measuring heading jitter and path consistency, and an 1800-frame accepted run for sustained system validation. A train/test split confirms generalization: videos used for fine-tuning (VID_017–020) are evaluated separately from held-out videos (IMG_1878, IMG_1921, IMG_1922, IMG_1924) (Chapter 4).
5. A GPS-conditioned waypoint-turn planner with a containment safety guard that verifies committed paths against the BEV corridor before execution, validated with a 0% containment failure rate (Chapter 3, Chapter 4).

1.5 Thesis Organization

Chapter 2 reviews prior work and identifies research gaps. Chapter 3 describes the complete system design. Chapter 4 presents experimental results. Chapter 5 discusses implications, limitations, and threats to validity. Chapter 6 summarizes contributions and outlines future work. Figure 1.1 shows the hardware platform.



(a) Scooter platform with mounted camera.



(b) Example first-person view from the camera.

Figure 1.1: Hardware platform. (a) Electric scooter used for data collection, with a forward-facing monocular camera rigidly mounted to the handlebar stem. (b) Sample sidewalk scene as seen by the camera; all experiments are conducted offline on video recorded from this rig.

CHAPTER 2

BACKGROUND AND RELATED WORK

This chapter surveys prior work in four areas that form the technical foundation for this thesis: sidewalk perception and navigation platforms, bird’s-eye view projection and path planning, distance-transform and skeleton-based path extraction, and semi-supervised training for embedded deployment. Each area reveals a specific gap that motivates the contributions of this thesis.

2.1 Sidewalk Perception and Navigation

Autonomous sidewalk navigation differs fundamentally from road-driving autonomy in sensor budget. Road vehicles employ LiDAR arrays and GPU clusters; sidewalk robots must achieve comparable reliability with cameras costing tens of dollars and single-board computers consuming under 15 watts. Starship Technologies has deployed thousands of delivery robots using cameras, ultrasonic sensors, and pre-mapped routes [1]. Viteri et al. [8] employ an RGB-D architecture for sidewalk-following. Machkour et al. [11] demonstrate monocular navigation for indoor corridors. Most platforms rely on depth sensors, multi-camera rigs, or pre-mapped routes—none provides a systematic comparison of planning strategies across coordinate domains for monocular sidewalk navigation.

Semantic segmentation of drivable surfaces underpins these systems. The architecture landscape ranges from FCN [12] and DeepLabV3+ [13] through lightweight models such as BiSeNetV2 [14] and DDRNet [15] to transformer-based designs. SegFormer [6] combines a hierarchical Vision Transformer encoder [16] with a lightweight MLP decoder, scaling from B0 (3.7M parameters, suitable for embedded deployment) to B5 (84.7M). Sidewalk-specific datasets remain limited: CityScapes [17] focuses on vehicle-mounted European street

scenes, RUGD [18] covers off-road terrain, and Mapillary Vistas [19] differs in viewpoint from low-mounted micro-mobility cameras. This scarcity motivates the semi-supervised teacher–student approach adopted in this thesis (Section 3.2.1).

2.2 BEV Projection and Path Planning

Bird’s-eye view representations provide a spatially consistent top-down reference frame for metric reasoning. The classical approach, Inverse Perspective Mapping (IPM) [20], uses a planar homography to warp the camera image onto the ground plane [21, 22]. IPM assumes flat ground and known extrinsics—assumptions that hold approximately for level sidewalks but degrade on ramps and uneven terrain. Learned BEV methods such as Lift-Splat-Shoot [3], BEVFormer [4], and Focus on BEV [5] relax the flat-ground assumption but require multi-camera rigs and GPU-class hardware that exceed the budget of a single-board ARM computer.

A less-studied limitation of monocular BEV is *computational cost*. A single camera maps only a narrow trapezoidal strip into the BEV grid, yet path *discovery* algorithms—skeletonization, Dijkstra search, distance-transform ridge extraction—that operate on this grid consume hundreds of milliseconds per frame. The cost–benefit trade-off of BEV path discovery versus BEV path verification has not been systematically characterized for sidewalk navigation. This thesis documents a two-orders-of-magnitude cost difference (734–927 ms vs. 3–5 ms) and demonstrates that verification retains the metric-scale benefits essential for turn geometry (Section 4.3).

Local path planning spans graph search [23, 24], potential fields [25], the Dynamic Window Approach (DWA) [26], trajectory optimization [27], and end-to-end learned policies [2, 28]. Agricultural row-following [9, 10] provides perhaps the closest analog to sidewalk corridor navigation, but sidewalks curve, narrow, and branch unpredictably. Most prior work plans in BEV or world-frame coordinates, implicitly assuming metric-scale

reasoning is necessary. No prior study has systematically compared BEV-domain and image-space planning for monocular sidewalk navigation—the comparison this thesis provides across ten methods in four algorithm families (Chapter 4).

2.3 Distance Transform and Skeleton Methods

The Euclidean distance transform (EDT) [29] computes per-pixel clearance from the nearest boundary, producing a clearance map whose ridge traces the corridor centerline. In 2D mask planning, the EDT ridge can be traced via Dijkstra search on a cost field $c(x, y) = 1/(d(x, y) + \epsilon)^\alpha$, naturally steering toward wide corridor centers. This thesis introduces an alternative use: employing the distance-transform corridor as a *verification surface* against which pre-computed template arcs are scored, shifting from discovery to approval (Chapter 3).

Morphological skeletonization [30, 31] reduces a binary mask to its medial axis, enabling graph-based path enumeration. However, skeletonization is noise-sensitive—small mask protrusions generate spurious branches requiring heuristic pruning—and the full pipeline (thinning, pruning, graph construction, Dijkstra search) carries substantial per-frame cost. No work in the surveyed literature has compared skeleton-graph construction against template-based verification for real sidewalk data; this thesis provides such a comparison (Chapter 4).

2.4 Semi-Supervised Learning and Embedded Deployment

Knowledge distillation [32] transfers capacity from a large teacher to a compact student by training on soft probability distributions rather than hard labels. Semi-supervised extensions use the teacher to generate pseudo-labels on unlabeled data [33]. OneFormer [7] unifies semantic, instance, and panoptic segmentation through task-conditioned joint training; pre-trained on ADE20K [34], COCO [35], and CityScapes [17] with a Swin-L backbone [36], it provides high-quality pseudo-labels across visual domains. The specific combination

of OneFormer Swin-L as teacher and SegFormer-B0 as student has not been explored for sidewalk binary segmentation; this thesis employs it to achieve IoU of 0.964 at 14.0 ms on CPU after fine-tuning on 228 human-corrected masks (Chapter 4).

Deploying on embedded hardware (Raspberry Pi 4, Rock 5B) requires careful attention to model selection and pipeline architecture. A 10 Hz (100 ms per frame) pipeline suffices for platforms moving at 1–2 m/s [37, 38]. Runtime optimization frameworks such as ONNX Runtime [39] and model quantization can reduce latency, but complete perception-to-planning pipelines are rarely benchmarked end-to-end on CPU-only hardware for sidewalk applications. This thesis presents such benchmarking, with the recommended image-space architecture achieving 50 FPS on CPU (Chapter 4).

2.5 Summary and Research Gap

The preceding sections reveal four research gaps:

1. **No systematic BEV-vs-image-space comparison for sidewalk planning.** This thesis provides a ten-method, two-domain benchmarking study (Chapter 4).
2. **BEV discovery-vs-verification cost–benefit uncharacterized.** The BEV cost–benefit analysis in Chapter 4 documents a two-orders-of-magnitude cost difference and demonstrates that verification retains metric-scale benefits for turn geometry.
3. **OneFormer-to-SegFormer distillation not applied to sidewalk segmentation.** This thesis presents such a recipe, achieving IoU of 0.964 at 14.0 ms on CPU.
4. **Few end-to-end CPU-only sidewalk pipeline benchmarks.** System-level profiling reports 50 FPS for the recommended image-space architecture (segmentation through path planning on CPU).

These gaps motivate the system design described in Chapter 3.

CHAPTER 3

SYSTEM DESIGN

The system evolved through four design iterations: (v1) BEV skeleton-graph planning, abandoned due to topological instability; (v2) BEV distance-transform ridge following, abandoned due to computational cost (~ 1 s per frame); (v3) image-space midpoint planning, which bypassed BEV entirely for straight-ahead following; and (v4) template-approval planning, which returned to BEV only for corridor verification and turn planning. Two domains coexist in the final system: BEV provides metric-scale coordinates for corridor extraction and turn planning, while image-space processing handles efficient straight-ahead path following. Figure 3.1 illustrates the overall architecture.

3.1 Hardware Platform

The hardware platform imposes the root constraint from which all subsequent design choices cascade. A single monocular camera operating under a 15 W power budget limits the model size, inference strategy, and sensing modality available to the system. The absence of a depth sensor, stereo camera, LiDAR, or IMU means that all three-dimensional reasoning must be derived from calibrated geometry—specifically, a planar homography—rather than from direct depth measurement.

The experimental platform is an electric scooter equipped with a single forward-facing monocular RGB camera (iPhone 15 rear camera, 24 mm equivalent focal length, f/1.6) rigidly mounted to the handlebar stem at approximately 0.8 m above ground level, angled approximately 10° downward from horizontal (Figure 1.1). The camera captures 1920×1080 video at 30 FPS in landscape orientation. No depth sensor, stereo camera, LiDAR, or IMU is used. The target deployment hardware is a Raspberry Pi 4 (Cortex-A72, 8 GB RAM)

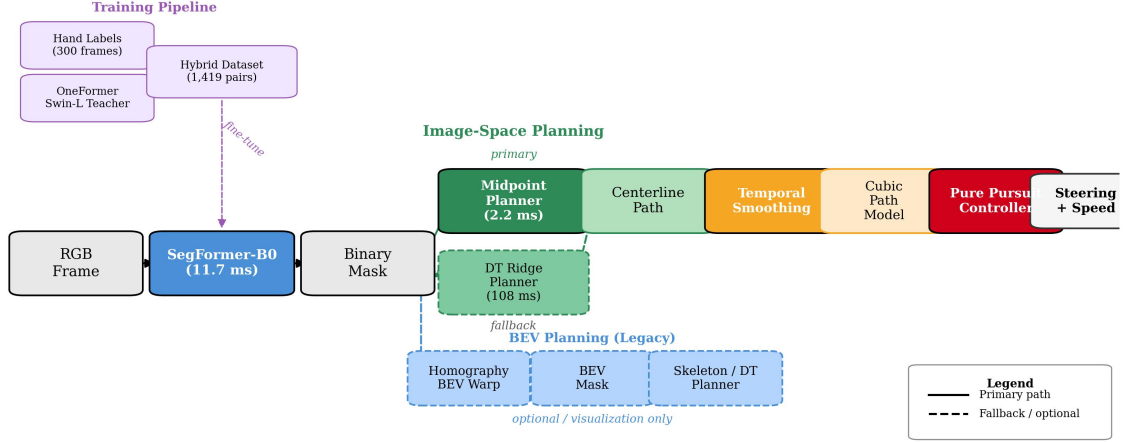


Figure 3.1: Overview of the sidewalk navigation pipeline. Hand-annotated and pseudo-labeled data are merged to train a student segmentation model, followed by geometric processing to extract traversable paths. The system supports both BEV-domain and image-space planning; the experiments in Chapter 4 compare ten methods across four algorithm families.

or Rock 5B class single-board computer; all development and evaluation were conducted on a workstation (Intel Core i7-12700K, 32 GB RAM, NVIDIA RTX 3060) with runtime measurements reported for CPU-only inference (GPU disabled) to approximate embedded deployment conditions. Actual Raspberry Pi latencies are expected to be $2\text{--}3\times$ higher than the reported workstation CPU times due to lower clock speeds and memory bandwidth; the 10 Hz target includes this margin.

These hardware constraints directly shape the segmentation strategy described next.

3.2 Segmentation Module and Supervision Strategy

The 10 Hz real-time constraint imposes a strict parameter budget: SegFormer-B0 [6] (3.7M parameters) achieves sub-50 ms inference on CPU, leaving sufficient budget for downstream processing. Larger variants (B2, B5) consume the entire per-frame budget on segmentation alone.

3.2.1 Teacher–Student Supervision

A semi-supervised teacher–student framework addresses the limited availability of sidewalk segmentation data. Two teacher architectures were compared: (1) a SegFormer-B2 teacher fine-tuned on an initial set of hand-annotated frames, and (2) a high-capacity OneFormer Swin-L teacher [7] pre-trained on ADE20K, generating pseudo-labels from 1,419 unlabeled frames across 10 campus videos. For each unlabeled image, the teacher produces pixel-wise probabilities thresholded at $\tau = 0.60$ to generate binary pseudo-labels, cleaned with connected-component filtering. The training set merges pseudo-labeled frames with hand-annotated frames (hand annotations take precedence where both exist).

3.2.2 Limitations of Teacher Pseudo-Labels

The teacher introduces systematic errors due to a **domain gap**: OneFormer is pre-trained on vehicle-mounted imagery (1.5–2.0 m camera height), while the scooter camera sits at 0.8 m, producing a substantially different perspective. Three failure categories emerged during a 228-frame annotation campaign on VID_017–020:

- **Unreachable surface segmentation:** The teacher segments any road-like surface—including background roads behind fences or elevated parallel paths—that is physically unreachable by the scooter.
- **Shadow and sunlight failures:** Strong shadows cause the teacher to miss shaded sidewalk or hallucinate road in shadowed grass; direct sunlight blends overexposed concrete with adjacent surfaces.
- **Seasonal appearance variation:** Dormant brown grass resembles concrete, causing boundary bleed the teacher cannot resolve.

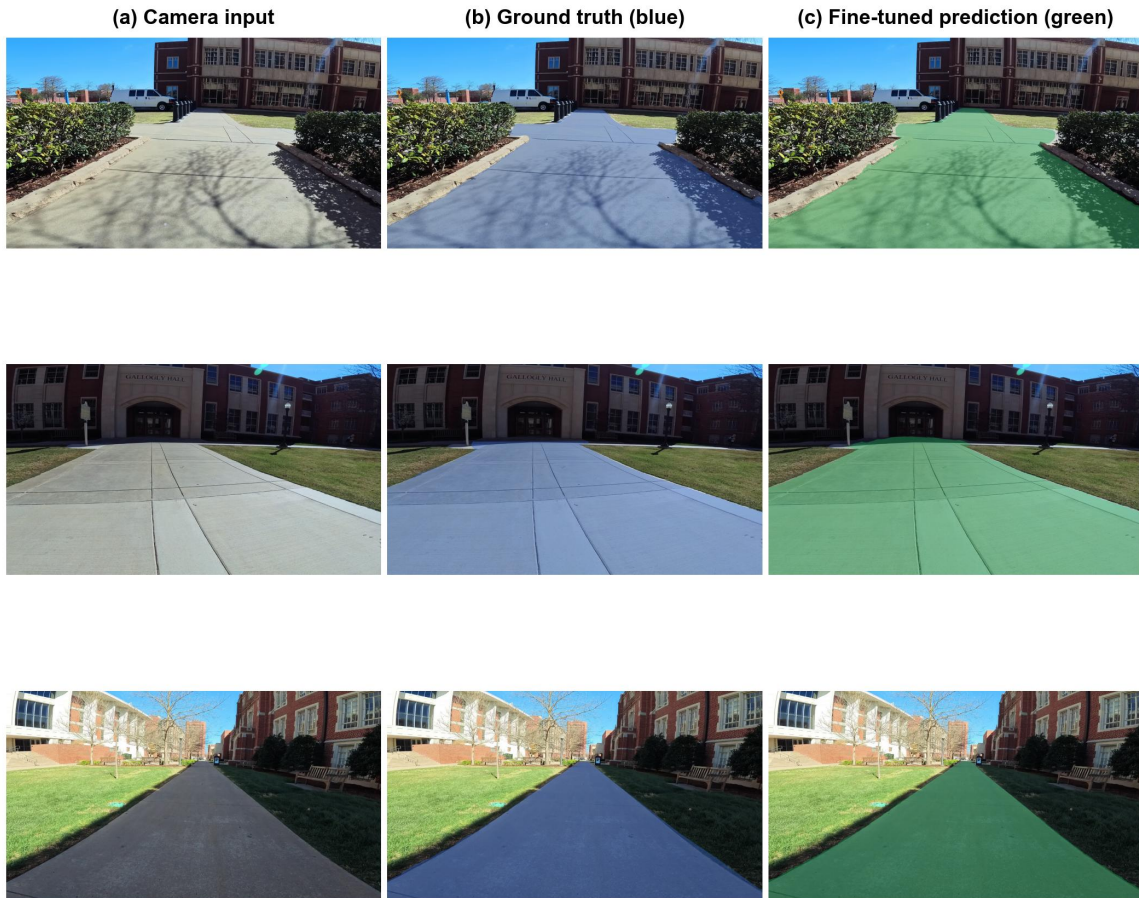


Figure 3.2: Segmentation quality on training videos (VID_017–019). (a) Camera input, (b) hand-corrected ground truth (blue overlay), (c) fine-tuned model prediction (green overlay). The fine-tuned model closely matches the ground truth, with tight sidewalk boundaries and minimal background over-segmentation.

3.2.3 Human-Corrected Fine-Tuning

The student model was fine-tuned on 228 hand-annotated frames using a strict annotation criterion: *traversable ego-reachable sidewalk only*. The recipe initializes from the teacher-trained checkpoint and trains for 8 epochs (lr 2×10^{-5} , cosine annealing, batch size 4, cross-entropy + Dice loss). Fine-tuning raises IoU from 0.934 to 0.964 and precision from 0.951 to 0.981 (Table 4.2), confirming that the model learned to suppress over-segmentation. The student minimizes a composite loss $\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}$ with class weights upweighting the minority sidewalk class.

Table 3.1 summarizes the full training progression.

Segmentation Quality: Baseline vs. Student Model

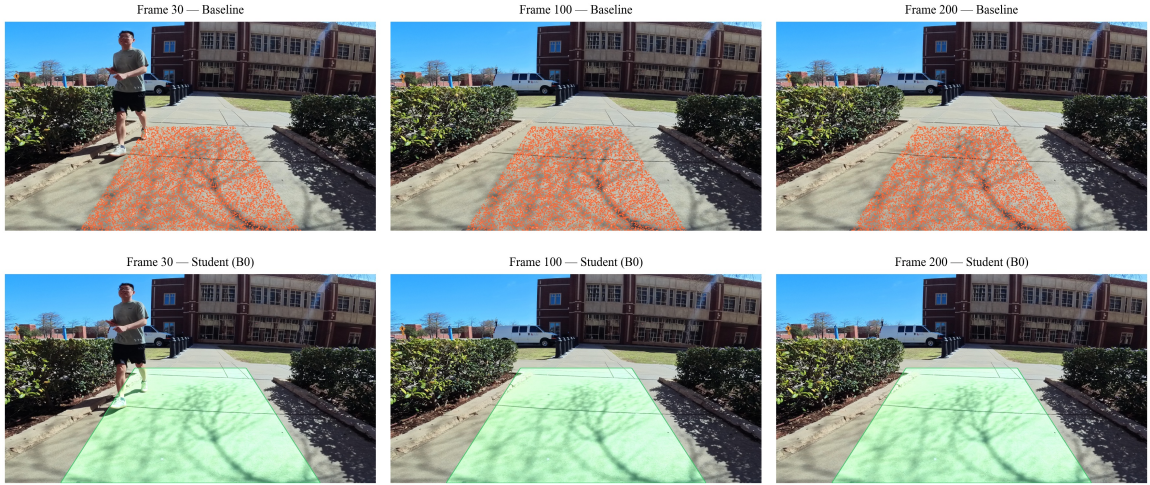


Figure 3.3: Qualitative segmentation comparison. Top: baseline (B2 teacher). Bottom: teacher-trained (OneFormer teacher). The OneFormer-trained model produces more complete, cleaner masks.

Table 3.1: Segmentation training progression across four stages. The student architecture (SegFormer-B0, 3.7M parameters) is fixed throughout; improvements arise from upgrading the teacher model and expanding the training data. Val IoU is measured on a held-out validation split; hand-annotated IoU (Table 4.2) uses an independent 228-frame evaluation set.

Stage	Teacher	Training Frames	Val IoU	Latency (ms)
1	SegFormer-B2 (300 hand-labeled)	699	0.880	18.9
2	OneFormer Swin-L (ADE20K)	699	0.944	11.7
3	OneFormer Swin-L + boundary loss	699	0.948	11.7
4	OneFormer Swin-L + expanded data	1,419	0.959	11.7

The interaction between input resolution and inference speed is characterized in the following section.

3.3 Resolution Trade-Off Analysis

Input resolution directly trades segmentation quality against inference latency, and in a real-time pipeline this trade-off propagates to every downstream module: a coarser mask degrades BEV corridor extraction, while a finer mask may consume the entire frame budget before path planning begins. Identifying the resolution sweet spot is therefore a prerequisite for the rest of the system design.

To balance accuracy and real-time performance, a resolution sweep using SegFormer-B0 on CPU characterizes the latency–quality trade-off. Table 3.2 reports latency and throughput. Lower resolutions (320×180 and below) produce masks that are overly coarse, fragmenting continuous sidewalk segments. Higher resolutions ($960 \times 540+$) incur prohibitive latency. The selected operating point is 640×360 : accurate segmentation at sub-50 ms inference. Figure 3.4 illustrates the visual quality at each resolution.

Table 3.2: SegFormer-B0 inference latency and throughput at various input resolutions on CPU. The 640×360 resolution provides the best balance of boundary detail and real-time feasibility, operating below 50 ms per frame while maintaining fine-grained mask edges.

Resolution	Latency (ms)	FPS
160×90	19.84	50.40
320×180	25.35	39.45
480×270	42.05	23.78
640×360	46.00	21.74
768×432	93.55	10.69
960×540	152.45	6.56
1280×720	352.40	2.84

With the segmentation resolution established, the next section describes how the resulting mask is projected into bird’s-eye view coordinates.

3.4 Bird’s-Eye View Projection and Mask Refinement

Corridor extraction, template-arc scoring, and turn planning require metric-scale coordinates. A four-point planar homography ($\mathbf{p}' \sim H\mathbf{p}$, where H is a 3×3 matrix estimated from four ground-plane correspondences) costs approximately 0.9 ms on CPU and maps the segmented mask to a BEV grid covering $6\text{m} \times 11\text{m}$ at 360×660 pixels (60 px/m). The homography assumes flat ground and a fixed camera pose—limitations quantified in Section 4.3. BEV is assigned to tasks where metric-scale reasoning is essential (corridor verification, turn geometry), while image-space handles straight-ahead path following.

The perspective warp amplifies segmentation noise at far distances. A multi-stage

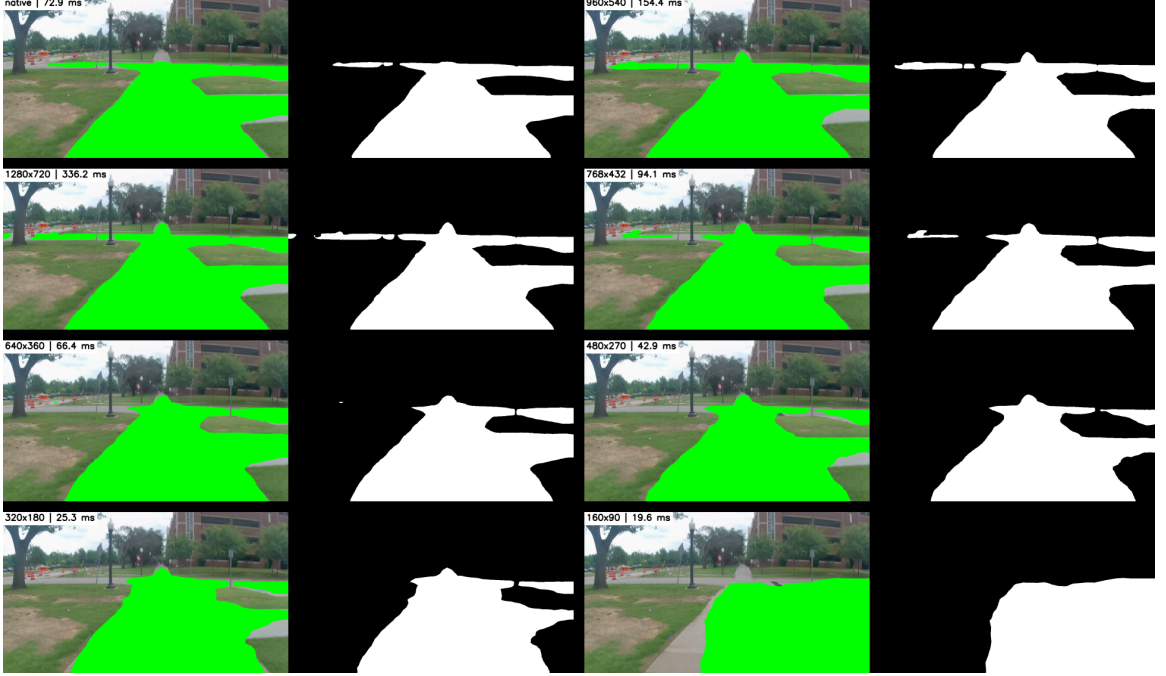


Figure 3.4: Resolution sweep for SegFormer-B0. Each column shows input overlay (left) and resulting segmentation mask (right). Higher resolutions improve edge detail but increase latency substantially.

cleanup pipeline applies morphological closing/opening, connected-component filtering (retaining only the ego-connected region), and optional enhanced refinement (flood-fill hole filling, Gaussian boundary smoothing, distance-transform-based component selection). The refined BEV mask and image-space mask serve as input to the ten planning methods described next.

3.5 Path Planning Methods

Ten path planning methods spanning two geometric domains and four algorithm families are compared. The central lesson of the design progression is that the planning *domain*—BEV versus image-space—matters as much as the planning *algorithm*. Table 3.3 provides a compact overview; the subsections below describe each method.

Table 3.3: Summary of the ten path planning methods evaluated in this thesis, grouped by algorithm family.

Method	Domain	Family	Strategy	Complexity
Midpoint	Image	Geometric	Per-row boundary averaging	$O(H \cdot W)$
Weighted Centroid	Image	Geometric	Area-weighted center of mass	$O(H \cdot W)$
Image DT	Image	Cost-field	EDT ridge + drift constraint	$O(H \cdot W)$
Image DT-Ridge	Image	Cost-field	EDT ridge tracing (image)	$O(H \cdot W)$
BEV DT-Ridge	BEV	Cost-field	EDT ridge tracing (BEV)	$O(N \log N)$
Skeleton-Graph	BEV	Topological	Graph search (Dijkstra)	$O(N \log N)$
Skeleton Hybrid	Image	Topological	Coarse skeleton + DT refine	$O(H \cdot W)$
Template-Approval	BEV	Template	Propose-and-verify (7 arcs)	$O(K \cdot L)$
Dynamic Window	Image	Sampling	Velocity-space sampling	$O(K \cdot T)$
Potential Field	Image	Sampling	Attractive/repulsive forces	$O(H \cdot W)$

3.5.1 BEV Skeleton-Graph Planner

The Guo–Hall parallel thinning algorithm [31] reduces the BEV mask to a 1-pixel-wide skeleton. An undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed from skeleton pixels with 8-neighbor edges, and Dijkstra’s algorithm [40] enumerates candidate paths scored by curvature and lateral shift. This method exposes junction topology but proved unstable in practice: minor mask perturbations create or destroy branches, causing path jumps between frames. At 734 ms per frame (Table 4.6), it consumes the entire real-time budget.

3.5.2 BEV Distance-Transform Ridge Planner

This method computes the EDT of the BEV mask and traces the ridge of maximum clearance via Dijkstra on a cost field $c(x, y) = 1/(d(x, y) + \varepsilon)^\alpha$ ($\varepsilon = 0.5$, $\alpha = 1.5$), smoothed with a Savitzky–Golay filter [41]. The ridge tracks the corridor center more consistently than skeletonization but inherits the full cost of BEV-domain graph search.

3.5.3 BEV Template-Approval Planner

A bank of seven pre-computed arc templates (straight plus gentle, medium, and sharp arcs in each direction) is scored against the distance-transform corridor [42]. Each template is evaluated by containment within the corridor, and the highest-scoring template passing a confidence gate becomes the planned path. Family-reuse hysteresis suppresses frame-to-frame switching. This approach eliminates noise-sensitive graph construction, reducing mean heading error by 40.6% and halving path-source switches compared to the skeleton-graph baseline (Table 4.9). This is the planner used in the 1,800-frame accepted run.

3.5.4 Image-Space Midpoint Planner

For each row y of the image-plane mask, the left and right sidewalk boundaries are identified and the midpoint $x_{\text{mid}}(y) = (x_L(y) + x_R(y))/2$ is computed. A connected-component filter ensures only the ego-connected region is used, and per-row midpoints are smoothed with a Savitzky–Golay filter. This method achieves 14.5 px lateral center error at 3.0 ms—the lowest error and fastest runtime among all ten methods—a $245\times$ speedup over BEV skeleton-graph planning. The central finding: for straight-ahead path extraction, the segmentation mask already encodes sufficient geometric information without BEV transformation.

3.5.5 Remaining Planners

Image-Space DT. The EDT is computed on the image-plane mask, and the maximum-clearance ridge is traced with a per-row lateral drift constraint. More robust than midpoint on irregular masks but slower (129.9 ms).

Image-Space DT-Ridge. Traces the global EDT ridge without drift constraint, allowing unconstrained ridge following (99.4 px error, 44.6 ms).

Weighted Centroid. Area-weighted center of mass per row, biased toward the corridor interior (51.1 px error, 2.4 ms—second-best accuracy).

Dynamic Window Approach. Samples admissible velocity pairs [26] and scores trajectories by heading alignment, clearance, and progress (77.0 px error, 3.6 ms). Exhibits the lowest heading jitter (2.1 deg/frame).

Potential Field. Attractive/repulsive forces [25] from goal and boundaries (154.4 px error, 2.8 ms). Suffers local-minimum sensitivity; paths on only 92% of frames.

Skeleton Hybrid. Coarse image-space medial-axis extraction refined by local EDT ridge snapping (82.1 px error, 8.1 ms—much faster than BEV skeleton-graph).

3.5.6 GPS-Conditioned Waypoint-Turn Planner

Turn maneuvers require metric-scale arc geometry that only BEV coordinates can provide. When a GPS waypoint triggers a junction maneuver, the planner operates in five stages: (1) scanning a forward decision band (2 m to 7 m) for asymmetric road openings on the commanded side; (2) clustering supported rows and selecting the candidate turn apex; (3) target gating with acquisition/sustain hysteresis (thresholds 0.40/0.25); (4) fitting a constant-curvature circular arc ($x = R \sin \phi$, $y = \pm R(1 - \cos \phi)$) from ego through the apex; and (5) containment gating requiring $\geq 60\%$ of path points inside the corridor and $\geq 70\%$ within the near-field (1.2 m).

3.5.7 Turn Containment Safety Guard

Every committed control path is evaluated against the BEV drivable mask before execution, measuring path-outside ratio, path-outside count, and minimum boundary clearance. The first 4 samples near the ego are excluded due to BEV bottom-edge noise. Paths failing

containment trigger hold behavior with speed reduction, ensuring unsafe paths never reach the controller. Overnight validation achieved a 0% containment failure rate.

3.6 Temporal Smoothing

Frame-by-frame inference produces unavoidable jitter. Three EMA filters are applied: (1) **path smoothing** on cubic polynomial coefficients with adaptive α (0.35–0.85) and topology-change reset; (2) **heading smoothing** with circular EMA handling the $\pm 180^\circ$ wraparound, resetting on jumps exceeding 45° ; and (3) **mask smoothing** with a conservative-blend policy ($\alpha = 0.65$, consistency threshold $c_{\text{thresh}} = 0.20$) that reduces blending when consecutive-frame IoU drops sharply.

3.7 Obstacle Detection, GPS Navigation, and Control

Obstacle detection. YOLOv8-nano [43, 44] (3.2M parameters, sub-30 ms on CPU) detects eight safety-relevant classes (person, bicycle, car, motorcycle, bus, truck, cat, dog) with confidence threshold $\tau_{\text{det}} = 0.35$. Distance is estimated from bounding-box foot position via a pinhole approximation ($d \approx h_c / \tan(\phi_v \cdot (y_{\text{foot}}/H - 0.5))$, $h_c = 0.8$ m). Obstacles within 1.0 m trigger a full stop; within 3.0 m, speed is reduced to 0.3 m/s.

GPS navigation. GPS serves as an intent-conditioning signal, not a localization source. A low-cost GNSS module (u-blox NEO-series, 1 Hz, 2.5 m CEP) provides waypoint intent: bearing difference below 12° yields straight intent, 12 – 40° yields left/right, above 40° yields sharp turn. The heading command blends vision ($w_v = 0.7$) and GPS ($w_g = 0.3$); when no fix is available, w_g falls to zero.

Steering and speed. A pure-pursuit-inspired control law [45] converts the heading angle to discrete commands: straight ($|\theta| < 12^\circ$, $v = 1.5$ m/s), left/right (12 – 40° , linearly interpo-

lated), or sharp turn ($> 40^\circ$, $v = 0.4$ m/s). Commands are transmitted via USB-serial with a 500 ms hardware watchdog.

3.8 Safety Mechanisms

The default response to uncertainty is to slow or stop [46]. Non-negotiable gates include: full stop with no valid path for N consecutive frames; immediate stop within 1.0 m of obstacles; GPS loss fallback to vision-only control; manual override via hardware button; and speed cap (0.2 m/s) when segmentation IoU drops below threshold for multiple consecutive frames.

3.9 Evaluation Metrics

The pipeline is evaluated offline using five metrics: (1) **Segmentation IoU** between predicted and ground-truth masks; (2) **Lateral center error** ($LCE = \frac{1}{N} \sum |x_{p_i} - c(y_i)|$, in pixels); (3) **Inside-GT ratio**—fraction of path points within the ground-truth mask; (4) **Temporal stability**—frame-to-frame mask IoU, with frames below 0.80 flagged as unstable; and (5) **Runtime** in milliseconds on CPU.

3.10 Software Architecture

The system is implemented in Python 3.10+ using OpenCV, PyTorch, NumPy, and SciPy. A centralized configuration module (`config.py`) serves as the single source of truth. Each research improvement is gated by a boolean flag, enabling exact reproduction of any prior iteration. Key reproducibility artifacts include the SegFormer-B0 checkpoints, BEV calibration matrix, 448 hand-annotated masks, and the centralized configuration.

CHAPTER 4

EXPERIMENTAL EVALUATION

This chapter presents experimental results organized around three themes: (1) segmentation quality and generalization, (2) the ten-method planner comparison with BEV cost-benefit analysis, and (3) system-level validation including the template planner, turn planner, and sustained operation. The experimental setup is described first.

4.1 Experimental Setup

All experiments use campus video collected at the University of Oklahoma during daytime hours (9 AM–4 PM) under clear to partly cloudy conditions. Eleven video sequences were recorded; six totaling 22 679 frames are used for full-video evaluation. A total of 448 hand-annotated frames across eight video sequences (VID_017–020, IMG_1878, IMG_1921, IMG_1922, IMG_1924) serves as the primary benchmark. Of these, 228 (VID_017–020) were used for fine-tuning and 220 (IMG_1878, IMG_1921, IMG_1922, IMG_1924) are held out. All annotations were produced by a single annotator (noted as a threat to validity in Section 5.4). Runtime is mean per-frame wall time on CPU (Intel Core i7-12700K, GPU disabled). Table 4.1 summarizes the dataset.

For the planner comparison, each method receives the same cleaned binary mask; BEV methods receive the homography-warped version. BEV planners run in stateless mode (fresh instance per frame). The fine-tuned model is trained on 228 frames from VID_017–020 with a 20% validation hold-out; the reported IoU (0.964) is the validation-set metric.

Table 4.1: Test video dataset summary. Six sequences totaling 22 679 frames are used for full-video evaluation.

Video ID	Approx. Frames	Conditions
IMG_1876	~2,000	Straight, mild shadows
IMG_1877	~2,500	Curves, tree canopy
IMG_1878	~3,000	T-junctions, surface changes
IMG_1921	~2,500	Wide path, open area
IMG_1922	~4,000	Mixed conditions
IMG_1924	~5,000	Diverse, pedestrian traffic
IMG_1931	~3,500	Narrow paths, tight turns
VID_017–020	~4,000 each	New campus routes, March 2026

4.2 Segmentation Quality and Generalization

Figure 4.1 illustrates the segmentation pipeline stages from RGB input through BEV projection. Table 4.2 compares the three-stage segmentation progression on 228 hand-annotated frames. Switching from a SegFormer-B2 teacher to OneFormer Swin-L raised IoU from 0.926 to 0.934; fine-tuning on human-corrected masks raised IoU to 0.964, driven by precision gains ($0.951 \rightarrow 0.981$) that confirm suppression of background over-segmentation. The student architecture (SegFormer-B0, 3.7M parameters) remained unchanged throughout all three stages; only the teacher, training data, and recipe varied. Per-video results (Table 4.3) confirm consistency across sequences.

Table 4.2: Segmentation quality progression on 228 hand-annotated frames (VID_017–020). Fine-tuning raises IoU by 3.0 points and precision by 3.0 points.

Model	IoU	Prec.	Recall	F1	ms
Baseline (B2 teacher)	0.926	0.946	0.978	0.960	15.6
Teacher-trained (OneFormer)	0.934	0.951	0.982	0.965	14.6
Fine-tuned (hand-annotated)	0.964	0.981	0.982	0.982	14.0

A full-video replay across six sequences (22 679 frames) confirms generalization: the candidate model reduces temporal instability by 77% ($1.46\% \rightarrow 0.33\%$) and increases template success rate by 5.6 percentage points (Table 4.4, Figure 4.4).

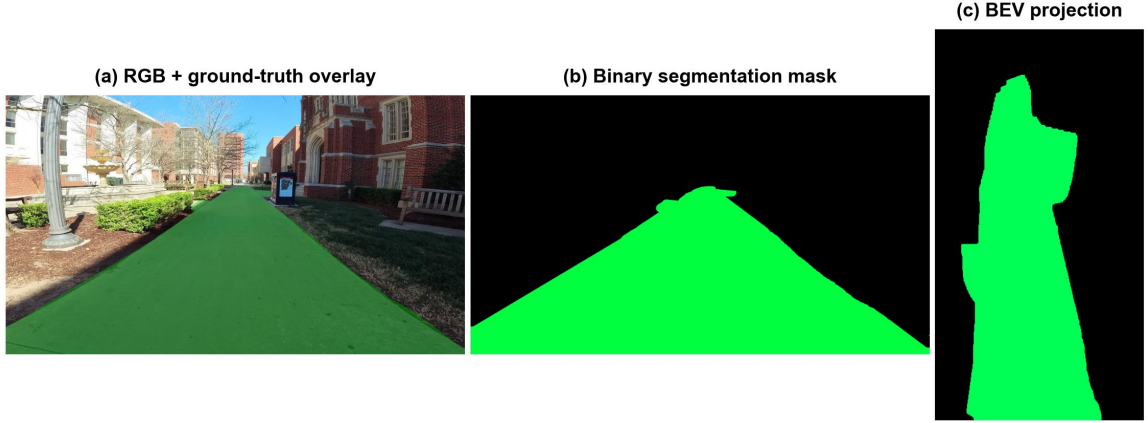


Figure 4.1: Segmentation pipeline stages on a campus sidewalk frame (VID_018). (a) RGB input with ground-truth mask overlay (green), (b) binary segmentation mask, (c) bird’s-eye view projection via calibrated homography.

Table 4.3: Per-video segmentation IoU for the fine-tuned model (228 frames).

Video	IoU	Precision	Recall
VID_017	0.971	0.982	0.989
VID_018	0.960	0.978	0.982
VID_019	0.962	0.988	0.974
VID_020	0.963	0.978	0.984
All	0.964	0.981	0.982

To verify that fine-tuning is not memorization, both models were evaluated on sequences *never used* during fine-tuning (IMG_1878, IMG_1921, IMG_1922). The fine-tuned model produces tighter masks with less background over-segmentation (Figure 4.5), and held-out IoU (0.974) exceeds training-set IoU (0.963), confirming generalization.

Figure 4.6 illustrates the complete perception-to-planning pipeline on three diverse campus scenes. Each row shows (a) the raw camera input, (b) the fine-tuned SegFormer prediction overlaid on the frame, and (c) planned navigation paths computed from the predicted mask, with a BEV inset showing the projected corridor.

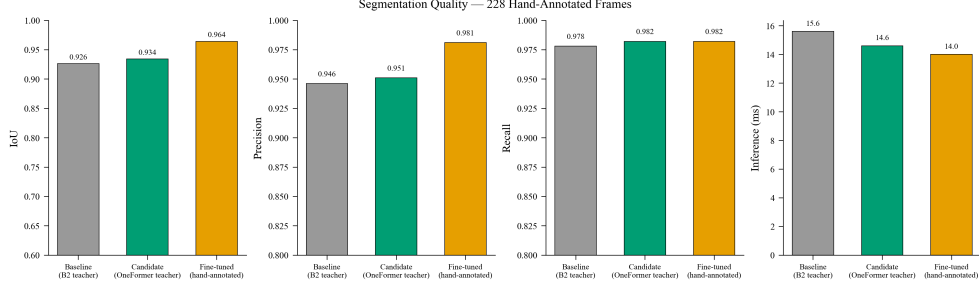


Figure 4.2: Segmentation quality comparison. The fine-tuned model achieves the highest IoU (0.964) by suppressing unreachable background surfaces.

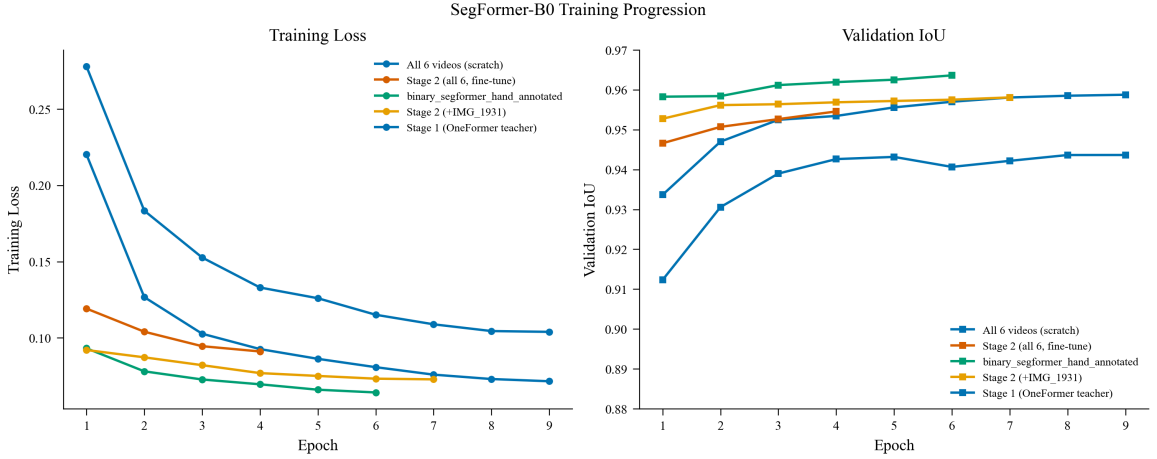


Figure 4.3: SegFormer-B0 training progression across four training runs. Left: training loss. Right: validation IoU. The stage 2 fine-tune on all six videos converges fastest due to checkpoint initialization.

4.3 Ten-Method Planner Comparison and BEV Cost-Benefit

The ten planners represent four design iterations (Table 4.5). Each iteration introduced new strategies while remaining backward-compatible. Ten methods are evaluated on 448 hand-annotated frames using oracle masks. The overall ranking is statistically significant (Friedman $\chi^2 = 311.1$, $p = 1.17 \times 10^{-61}$, $N = 71$ common frames across all ten planners; 32 of 45 pairwise Wilcoxon signed-rank tests significant after Bonferroni correction, including all nine comparisons involving the midpoint planner). The low N reflects the BEV DT-ridge planner’s 16% path-production rate; excluding it yields a nine-planner Friedman test with $\chi^2 = 1513.7$, $p < 10^{-318}$, $N = 410$.

Table 4.4: Full-video replay metrics across six sequences (22 679 frames). The candidate reduces temporal instability by 77%.

Metric	Baseline	Candidate	Δ
Mean Seg IoU	0.909	0.925	+0.016
Unstable Rate [%]	1.46	0.33	−1.13 pp
Template Success [%]	73.7	79.3	+5.6 pp
Fallback Rate [%]	19.0	14.3	−4.7 pp

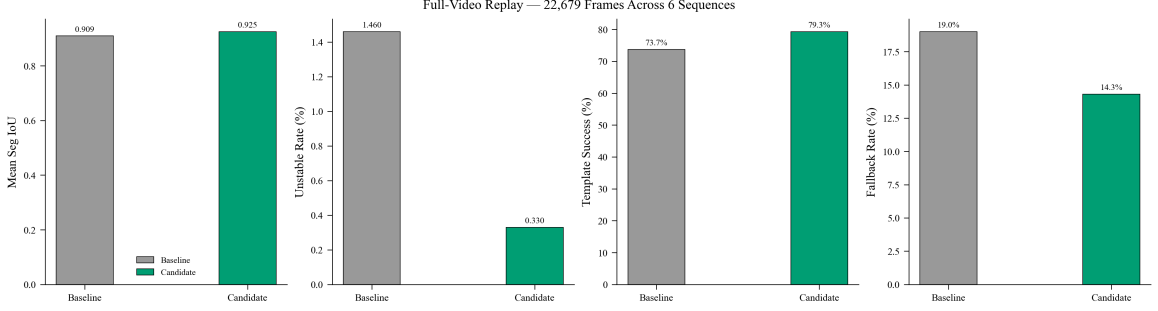


Figure 4.4: Full-video replay comparison across six sequences (22 679 frames). The candidate model reduces temporal instability by 77% and increases template success rate by 5.6 percentage points.

Table 4.6 reports accuracy and runtime; Table 4.7 isolates the effect of segmentation model on midpoint accuracy.

Figure 4.8 visualizes the accuracy–speed tradeoff. Midpoint occupies the Pareto-optimal corner (lowest error, fastest runtime). Four planners meet the 100 ms real-time budget: midpoint, weighted centroid, dynamic window, and potential field. Among these, midpoint dominates on accuracy while weighted centroid offers the second-best error (51.1 px).

The image-space midpoint planner achieves the lowest lateral center error (14.5 px, 95% CI [13.5, 15.4]) at the fastest runtime (3.0 ms). Fine-tuning reduces midpoint error from 35.6 px to 18.3 px (Table 4.7)—a 49% improvement approaching the oracle bound. The midpoint planner is intent-agnostic: it cannot execute commanded turns, motivating the dual-domain architecture. Among other planners, weighted centroid (51.1 px, 2.4 ms) and DWA (77.0 px, 3.6 ms) offer favorable tradeoffs within the real-time budget. Figure 4.9 summarizes these results; Figure 4.10 illustrates the BEV skeleton-graph pipeline.

Table 4.5: Design iteration progression. The shift from BEV discovery (v1–v2) to image-space (v3) yields a $25\times$ throughput improvement; v4 returns to BEV only for verification.

Iter.	Planner	Domain	Key Change	FPS
v1	Skeleton-Graph	BEV	Guo–Hall + Dijkstra	9.1
v2	DT Ridge	BEV	EDT ridge replaces skeleton	~ 1
v3	Midpoint / DT	Image-space	BEV bypass	59.2
v4	Template Arc	BEV corridor	Propose-and-verify	9.97

Segmentation Error Cascade

When planners receive predicted rather than oracle masks, midpoint degrades by only 3.8 px (14.5 px $\rightarrow$ 18.3 px), confirming robustness. Held-out segmentation IoU (0.974) exceeds training-set IoU (0.963), and midpoint lateral error on held-out videos (14.0 px) is lower than on training videos (22.4 px), reflecting easier geometry rather than overfitting. Figure 4.11 summarizes both the error cascade and the generalization evidence; Figure 4.12 shows the visual effect on three representative frames.

Intent-Conditioned Evaluation

Navigation intent labels (straight, left, right) were assigned by a human annotator who reviewed each video sequence and marked the frame ranges where the scooter’s physical heading changed direction during data collection. On 17 930 frames annotated with navigation intent, the midpoint planner maintains low center error (23–28 px) but achieves only 33–46% turn success, confirming it is intent-agnostic (Figure 4.14). No single-frame planner achieves reliable turn execution, confirming that the waypoint-turn planner (Section 4.5.2) is necessary.

Temporal Stability

The DWA exhibits the lowest heading jitter (2.1 deg/frame). Midpoint shows moderate jitter (4.5 deg/frame, $p_{95} = 18.5$ deg), motivating the temporal smoother. Right turns destabilize all planners $2\text{--}3\times$ relative to straight segments.

Table 4.6: Ten-planner comparison on 448 hand-annotated frames using oracle (ground-truth) masks. Planners are grouped by algorithm family. Runtime is mean per-frame wall time on CPU. The image-space midpoint planner achieves the lowest lateral center error (14.5 px) at the fastest runtime (3.0 ms). The BEV DT-ridge planner produces paths on only 16% of frames.

Family	Planner	Path [%]	Inside-GT	Err. [px]	ms
<i>Geometric (boundary-based)</i>					
	Midpoint	100	0.99	14.5	3.0
	Weighted Centroid	100	0.99	51.1	2.4
<i>Cost-field (distance-transform)</i>					
	Image DT	100	1.00	87.0	129.9
	Image DT-Ridge	100	1.00	99.4	44.6
	BEV DT-Ridge	16	0.80	48.0	174.2
<i>Topological (skeleton/graph)</i>					
	BEV Skeleton-Graph	100	0.96	84.1	734
	Skeleton Hybrid	100	0.99	82.1	8.1
<i>Template and sampling</i>					
	BEV Template-Approval	100	0.98	84.8	259
	Dynamic Window	100	0.99	77.0	3.6
	Potential Field	92	0.91	154.4	2.8

4.3.1 BEV Cost–Benefit Analysis

BEV masks achieve approximately 33% occupancy—adequate for corridor extraction and turn geometry. The bottleneck is computational: BEV path *discovery* costs 734–927 ms per frame, while BEV path *verification* costs only 3–5 ms (Table 4.8). Even with oracle masks, BEV skeleton-graph achieves 84.1 px error compared to 14.5 px for midpoint, confirming that the coordinate transform itself introduces error.

The dual-domain architecture resolves this: image-space for straight-ahead following, BEV for corridor verification and turn geometry where metric-scale reasoning is essential.

4.4 Template-Approval vs. Skeleton Discovery

Both planners are compared on a 220-frame calibrated clip (VID_017, detection disabled). Table 4.9 shows that the template planner reduces mean heading error by 40.6% and

Table 4.7: Effect of segmentation model on image-space midpoint planner accuracy (448 frames). Hand-annotated fine-tuning reduces lateral center error, approaching oracle performance.

Segmentation Model	Inside-GT	Center Err. [px]	ms
Baseline (old model)	0.987	39.4	3.0
Teacher-trained	0.968	35.6	3.0
Fine-tuned	0.985	18.3	3.0
Oracle (ground truth)	0.994	14.5	3.0

Table 4.8: BEV cost–benefit across design iterations. Discovery costs 734–927 ms; verification retains metric-scale benefits at 3–5 ms.

Iteration	BEV Role	Planner Cost	BEV Value
v1 (Skeleton-graph)	Path discovery	734 ms	Metric paths (slow)
v2 (DT-ridge corridor)	Path discovery	926.8 ms	Metric paths (slower)
v3 (Image-space)	None	2.2 ms	—
v4 (Template-approval)	Verification	3 ––5 ms	Corridor + turn geometry

halves path-source switches, confirming that the fixed arc bank with family-reuse hysteresis produces smoother control output than per-frame graph search. The lower mean speed (1.067 vs. 1.500 m/s) reflects confidence-modulated slowdown—a deliberate safety trade-off.

Table 4.9: Template-approval vs. skeleton-graph baseline (220 frames, VID_017). Template reduces heading error by 40.6% and halves path-source switches.

Metric	Graph Baseline	Template-Approval
Mean $ \theta $ [deg]	1.190	0.707
$p_{95} \theta $ [deg]	3.724	3.508
Path-source switches	36	18
Template rate [%]	—	62.3
Graph rate [%]	75.0	5.9
Fallback rate [%]	25.0	31.8
Mean speed [m/s]	1.500	1.067

4.5 System Validation

4.5.1 Runtime Analysis

Table 4.10 compares per-module runtimes. The planner dominates the difference: image-space midpoint costs 3.0 ms vs. 734 ms for BEV skeleton-graph, a $245\times$ speedup.

Table 4.10: Per-module runtime comparison at 640×360 on CPU. The planner module dominates: $245\times$ speedup for image-space.

Module	BEV Pipeline	Image-Space	Note
SegFormer Inference	14.0 ms	14.0 ms	Shared (fine-tuned)
Mask Refinement	8.5 ms	3.0 ms	Img skips BEV cleanup
BEV Projection	0.9 ms	—	
BEV Cleanup	14.6 ms	—	
Planner	734 ms	3.0 ms	$245\times$ speedup
Total	772 ms	20.0 ms	
FPS	1.3	50.0	

Table 4.11 shows profiled runtimes under different system configurations, illustrating the impact of detection mode and predictive frame reuse on throughput.

Table 4.11: System-level runtime under different configurations (4,407-frame run, 512×288 , CPU-only).

Configuration	FPS	Seg (ms)	Det (ms)	BEV (ms)
No detection, predictor on	25.3	20.3	—	11.0
GPU detection, predictor on	20.1	17.0	12.2	13.4
CPU detection, predictor on	12.2	18.8	39.0	14.6
No detection, predictor off	8.9	75.3	—	29.5

4.5.2 Waypoint-Turn Planner Validation

The turn planner is evaluated on a 300-frame replay of VID_017 with right intent commanded on frames 20–30. In the deployed system, GPS bearing comparison generates intent automatically; for this evaluation, intent was manually triggered at the actual turn locations

where the scooter physically changed direction during data collection, testing the complete pipeline (corridor scan, arc fitting, containment gating, template-to-turn-to-template transitions) on real video at real turn geometry. Table 4.12 confirms a 0% containment failure rate. The planner activates the turn path on frames 24–25 (corridor support exceeds the 0.40 acquisition gate) and maintains hold behavior on surrounding frames; outside the intent window, the system cleanly returns to template following. End-to-end validation with live GPS-triggered turns remains future work.

Table 4.12: Waypoint-turn planner evaluation (300 frames, VID_017). The containment guard achieves 0% failure rate.

Metric	Value
Total frames	300
Template rate [%]	96.3 (289/300)
Turn containment failure rate	0.0%
Min boundary clearance [px]	4.775
Mean FPS	12.86

4.5.3 1800-Frame Accepted Run

Table 4.13 summarizes the full-length evaluation on VID_017 (stride 4, detection disabled). The template planner achieves 100% success across all 1800 frames—a substantial improvement over the 62.3% template rate in the 220-frame evaluation (Table 4.9), attributable to the improved segmentation model (IoU 0.954) and stride-4 temporal sampling. The 9.97 FPS throughput meets the 10 Hz real-time requirement, confirming deployment viability on CPU hardware.

4.5.4 Temporal Smoothing Evaluation

On a 500-frame window, temporal smoothing reduces mean heading jitter by 66% ($3.2^\circ \rightarrow 1.1^\circ$ per frame) and path-source switches from 8 to 3 (Table 4.14). The smoother’s primary value is suppressing residual jitter; the reset hysteresis triggers only on genuine scene

Table 4.13: Accepted run on VID_017: 1800 frames, template-approval sustains 100% path availability at 9.97 FPS.

Metric	Value
Frames processed	1800
Mean loop FPS	9.97
Mean seg IoU	0.954
Has-path rate	100%
Path source	template (all 1800)
Mean pipeline time	69.5 ms

changes.

Table 4.14: Temporal smoothing effect (500-frame window). Heading jitter reduced by 66%.

Metric	No Smoothing	With EMA
Temporally stable frames (%)	97.8	99.6
Mean heading jitter (deg/frame)	3.2	1.1
Path-source switches	8	3

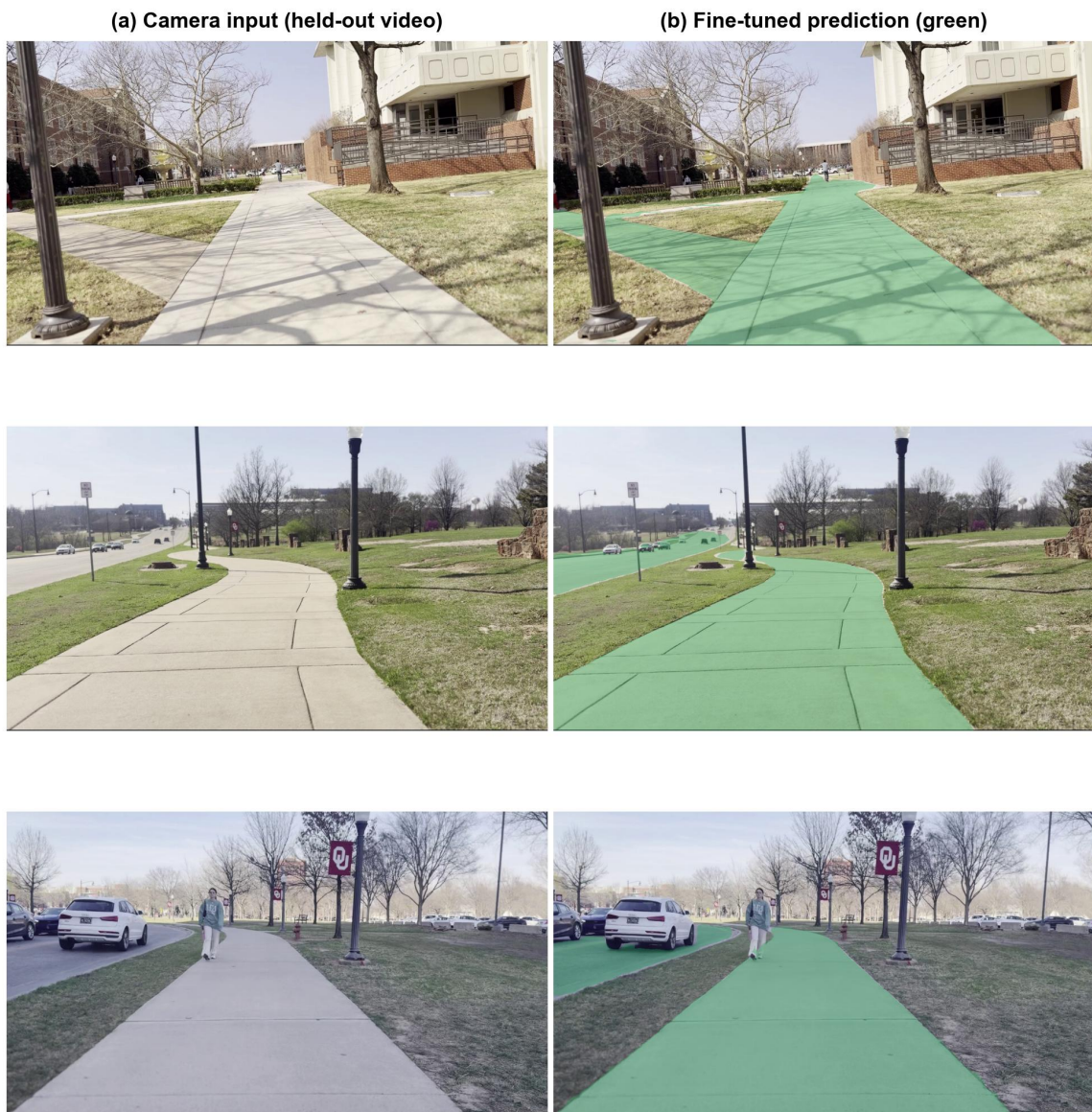


Figure 4.5: Segmentation on sequences *not used for fine-tuning* (IMG_1878, IMG_1921, IMG_1922). (a) Camera input from held-out videos, (b) fine-tuned model prediction (green overlay). The model generalizes to unseen campus locations, producing tight sidewalk masks without background over-segmentation (held-out IoU 0.974).

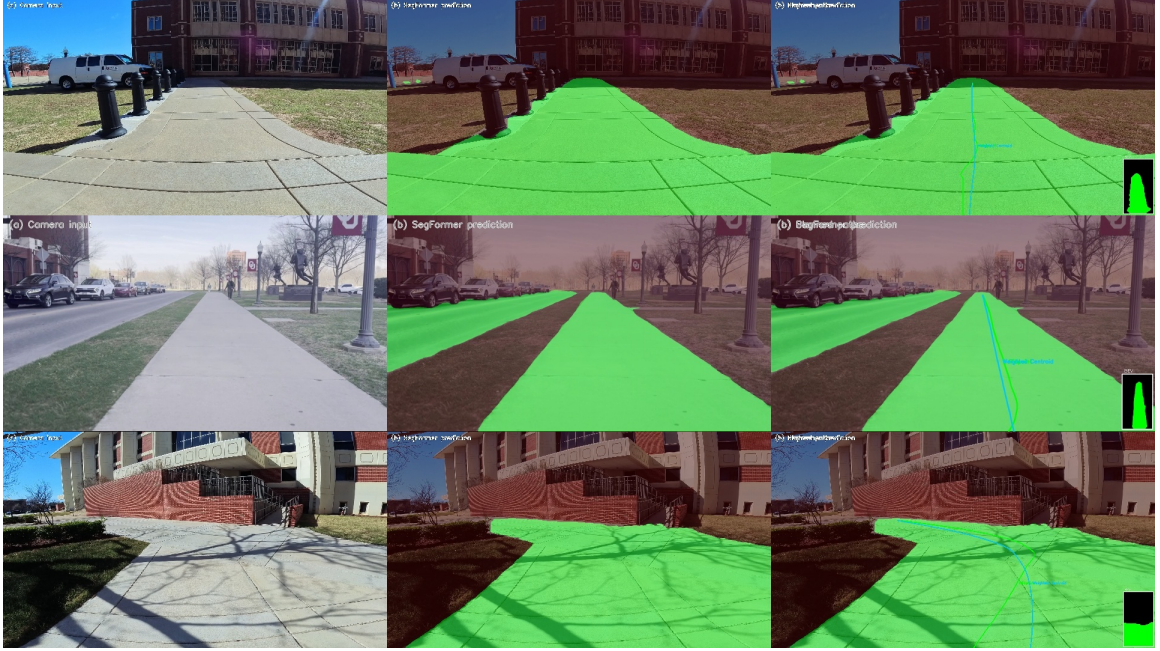


Figure 4.6: End-to-end pipeline demonstration on three campus sidewalk scenes (VID_017, IMG_1922, VID_018). Each row: (a) camera input, (b) fine-tuned SegFormer prediction (green overlay), (c) planned paths (green = midpoint, yellow = weighted centroid) with BEV corridor inset. The model produces clean masks across diverse geometry; both planners generate centered paths within the predicted corridor.

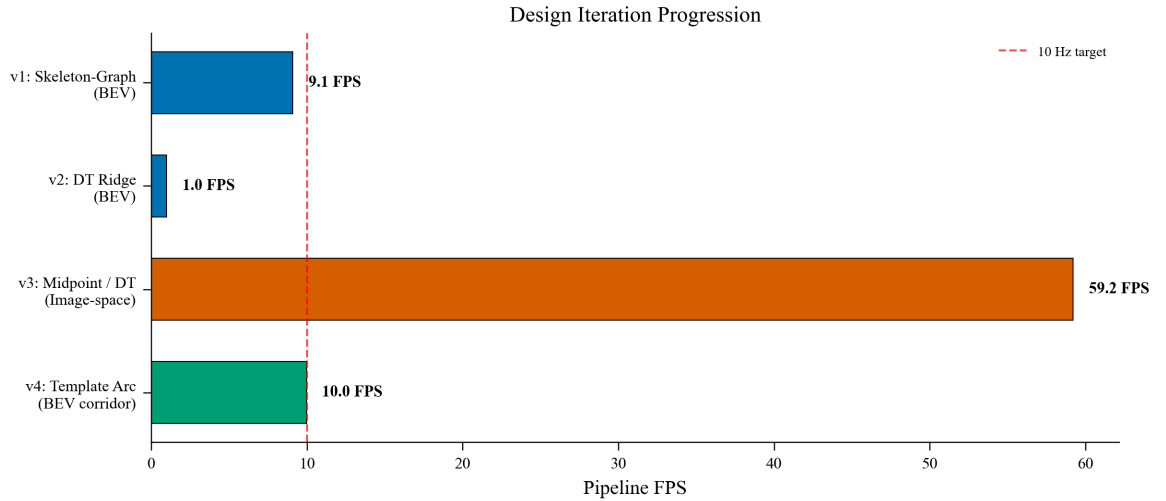


Figure 4.7: Pipeline throughput across four design iterations. The shift from BEV discovery (v1–v2) to image-space (v3) yields a $6.5\times$ improvement; v4 returns to BEV for verification at 10 Hz.

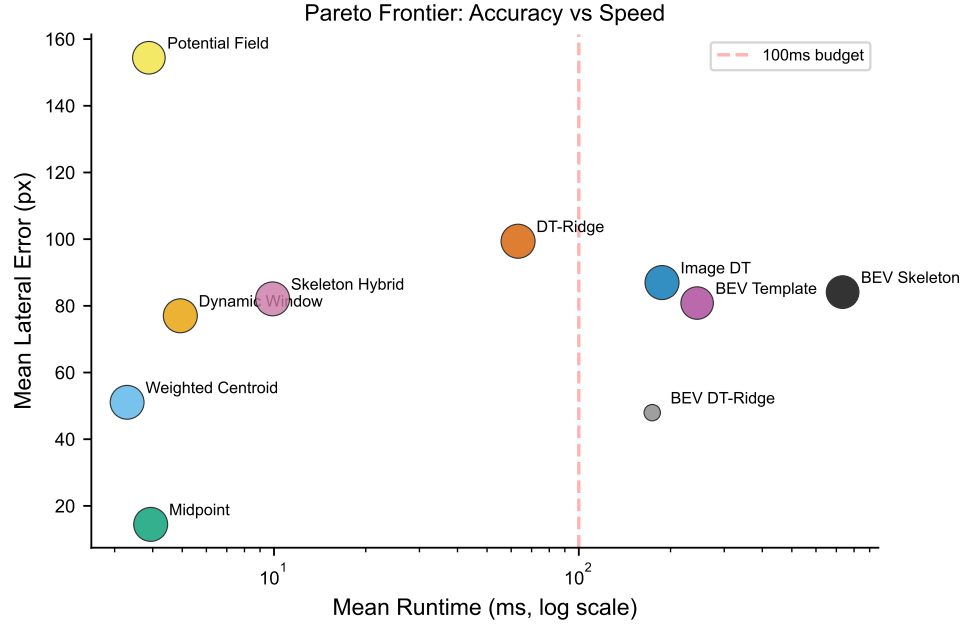


Figure 4.8: Pareto frontier of planning accuracy versus computational cost on 448 oracle-mask frames. Image-space midpoint (green, lower-left) dominates all other methods. The 100 ms real-time budget (red dashed line) divides viable from non-viable planners for 10 Hz deployment.

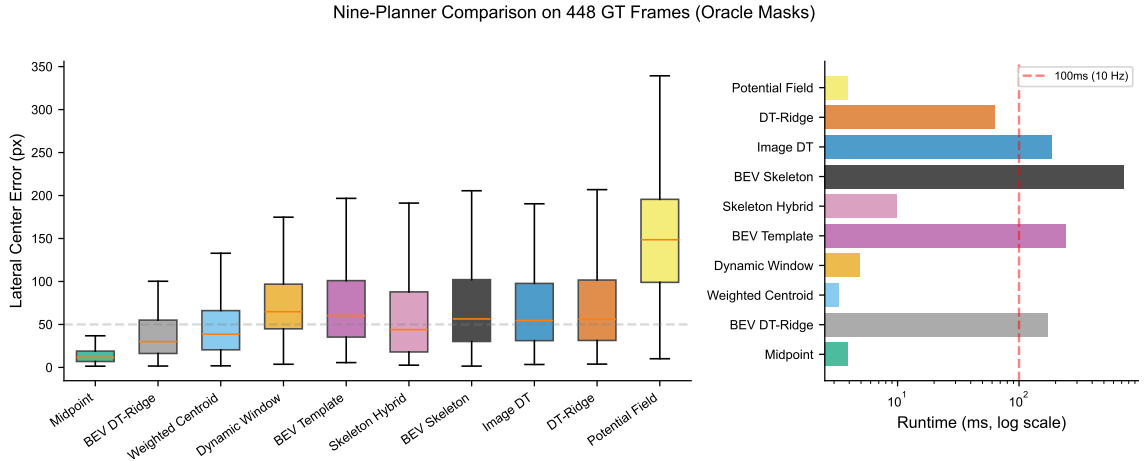


Figure 4.9: Ten-planner comparison on 448 hand-annotated frames. Left: lateral center error distribution (box plot). Right: per-planner runtime on logarithmic scale; the red dashed line marks the 100 ms real-time budget. Midpoint achieves the lowest error with the fastest runtime.

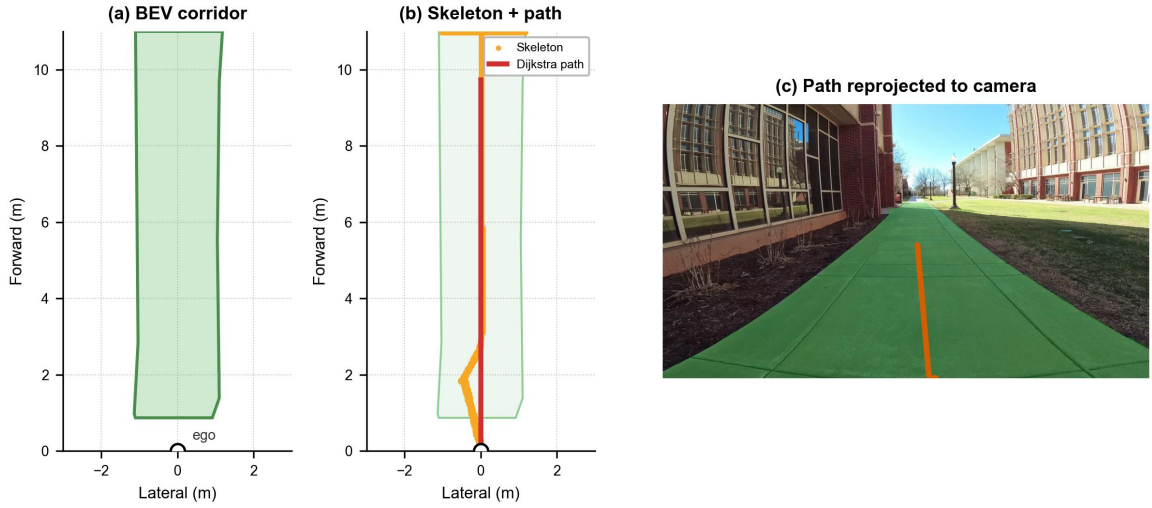


Figure 4.10: BEV skeleton-graph pipeline on a campus sidewalk frame (VID_017). (a) BEV mask after homography projection, (b) Guo–Hall thinning produces the skeleton (red), (c) selected Dijkstra path reprojected to camera view (red line on green mask overlay). While topologically informative, this method is the slowest (734 ms) and least accurate of the compared planners.

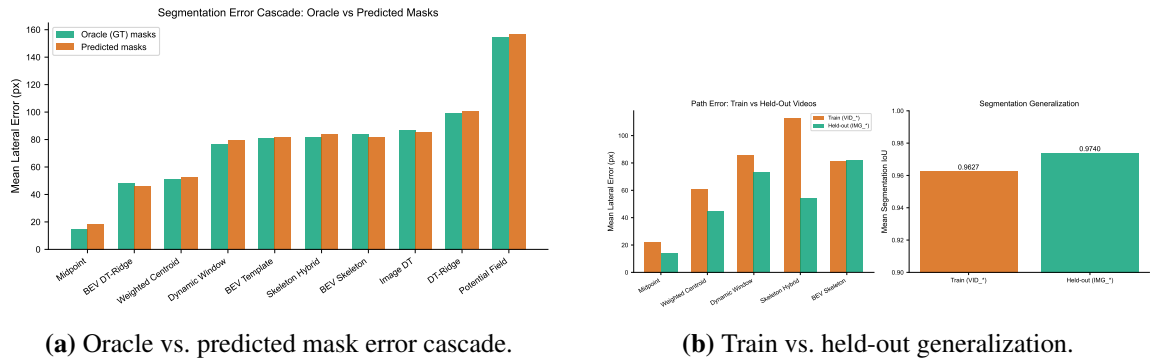


Figure 4.11: Segmentation robustness. (a) Most planners degrade by 1–4 px when switching from oracle to predicted masks. (b) Held-out IoU (0.974) exceeds training IoU (0.963), confirming generalization.

Segmentation Error Cascade: Oracle vs Predicted Mask Paths
 Green overlay = road mask. Colored lines = midpoint (green) and weighted centroid (yellow) paths.

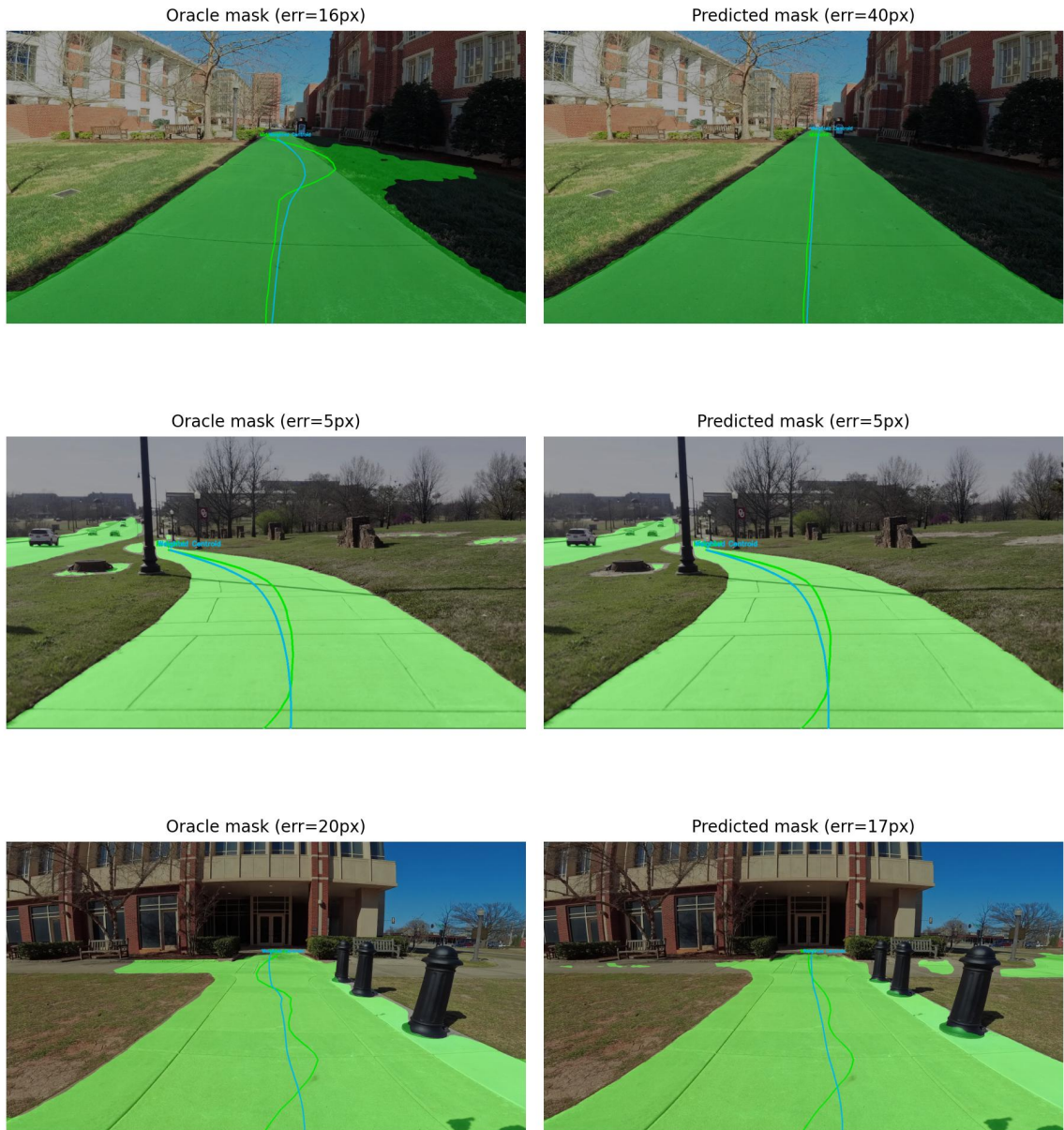


Figure 4.12: Visual comparison of planner paths on oracle (ground-truth) masks versus predicted masks for three campus sidewalk frames (VID_019, IMG_1921, VID_020). Green overlay = road mask. Green line = midpoint path, yellow line = weighted centroid path. Top row: predicted mask over-segments grass, shifting the midpoint path (16 px → 40 px error). Middle and bottom rows: predicted mask closely matches oracle, producing nearly identical paths.

Planner path comparison on campus sidewalk (IMG_1921)



Figure 4.13: Qualitative planner comparison on a curved campus sidewalk (IMG_1921). Green overlay = ground-truth road mask. Colored paths show four representative planners: midpoint (green), weighted centroid (orange), dynamic window (sky blue), BEV skeleton-graph (vermillion). The curve reveals path divergence—midpoint and weighted centroid track the corridor center, while skeleton-graph deviates laterally.

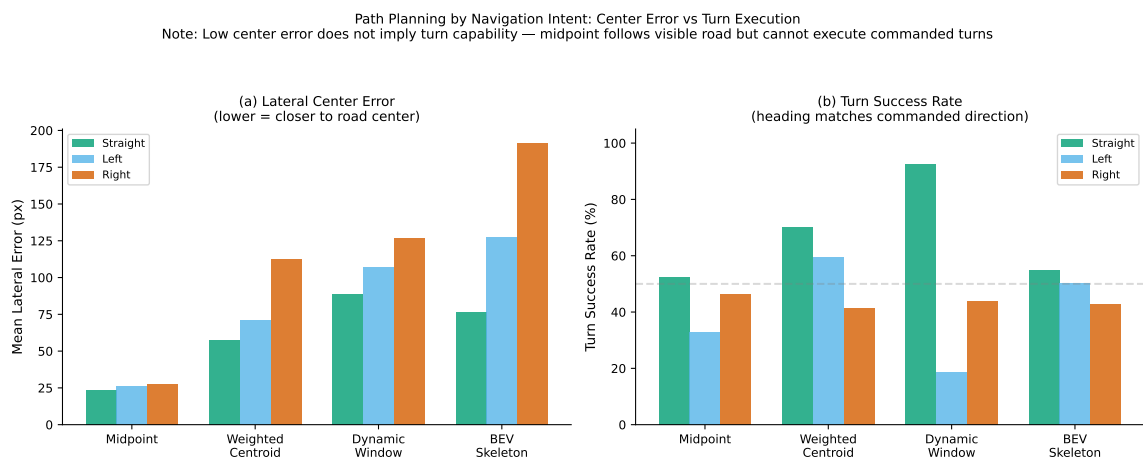


Figure 4.14: Planner performance by navigation intent. (a) Lateral error by intent. (b) Turn success rate: midpoint matches commanded direction only 33–46% of the time.

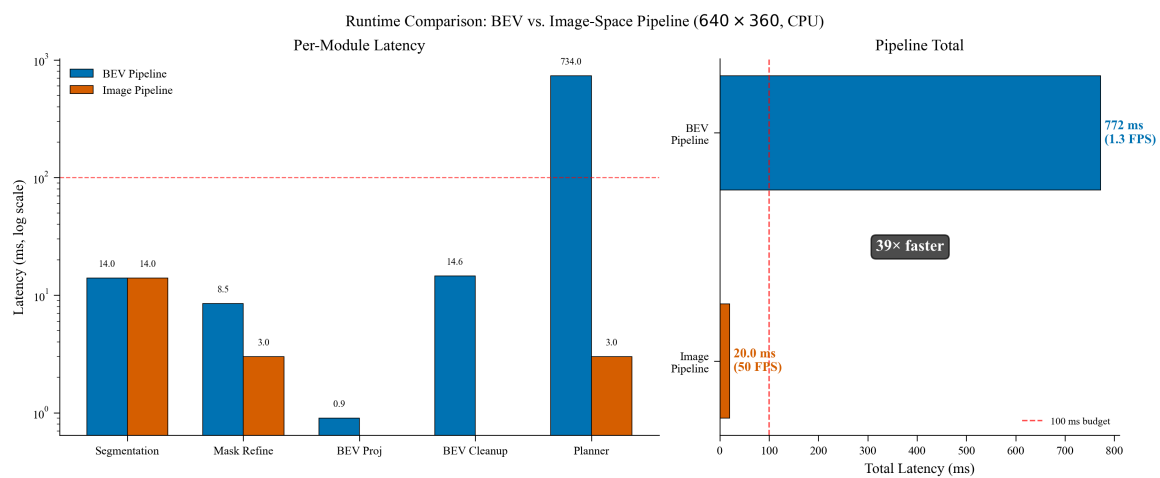


Figure 4.15: Per-module runtime comparison. Image-space achieves 50.0 FPS vs. 1.3 FPS for BEV skeleton-graph.

CHAPTER 5

DISCUSSION

5.1 Why Image-Space Outperforms BEV for Straight-Ahead Following

The central finding is that the optimal planning strategy depends on the navigation task. Image-space midpoint planning achieves 14.5 px lateral error at 3.0 ms—the lowest error and fastest runtime among all ten methods (Friedman $p < 10^{-61}$). This advantage persists with predicted masks (18.3 px) and on held-out videos (14.0 px). However, midpoint is intent-agnostic (33–46% turn success), motivating the dual-domain architecture.

The geometric explanation is straightforward. The segmentation mask fills 20–60% of image pixels, providing dense boundary information for per-row midpoint extraction without coordinate transformation. BEV projection concentrates near-field pixels into a dense region while stretching far-field pixels, producing uneven spatial resolution. Even with oracle masks, BEV skeleton-graph achieves 84.1 px error vs. 14.5 px for midpoint—the domain itself introduces error. The computational cost compounds the problem: path discovery algorithms on BEV grids require 734–927 ms per frame, while template verification costs only 3–5 ms.

Better segmentation helps all planners proportionally: fine-tuning reduces midpoint error by 49% (35.6 px $\rightarrow$ 18.3 px). However, BEV planning fails even with oracle masks, indicating that the bottleneck is coordinate-transform error and algorithmic cost rather than mask quality.

The template-approval planner addresses skeleton failures (topological instability, 734 ms cost, unnecessary generality) by replacing per-frame discovery with fixed-arc verification,

yielding 40.6% lower heading error and 50% fewer path-source switches. For turn handling, GPS intent conditioning separates *where to turn* (GPS bearing) from *how to turn safely* (BEV corridor verification), with a 0% containment failure rate.

5.2 Failure Modes

On 448 frames, 95.2% achieve center error below 60 px and 98.7% below 100 px. Four failure scenarios account for the remainder: (1) **Sharp turns** ($> 25^\circ$ heading) cause per-row midpoint drift—18.9% of frames exhibit this, motivating BEV turn planning. (2) **Narrow sidewalks** (< 1 m) reduce the pixel budget for boundary detection. (3) **Heavy occlusion** ($> 60\%$ mask loss) fragments the path; the temporal smoother bridges brief events while the stability gate caps speed for sustained loss. (4) **Far-range distortion** (> 8 m) degrades midpoint precision in the lookahead region, addressed by the 8 m planning horizon.

5.3 Broader Implications

The geometric coverage insight—image-space preserves denser planning information than BEV for monocular systems—may extend to agricultural row following [47], indoor hallway navigation, and trail following, though cross-domain evaluation is needed. The template-approval paradigm (propose-and-verify rather than discover) trades global optimality for computational efficiency and temporal stability—properties valuable on platforms with limited processing budgets. The 50 FPS CPU-only throughput for the image-space perception-to-planning pipeline demonstrates that sidewalk navigation is achievable without GPU acceleration.

5.4 Limitations and Threats to Validity

Offline evaluation. Paths are never executed on the physical scooter; metrics proxy but do not guarantee navigation safety.

Single annotator, single campus. All 448 annotations were produced by one annotator; no inter-annotator agreement is measured. All data comes from the University of Oklahoma campus during daytime conditions.

Fixed camera geometry. A single mount position and angle; different configurations would require recalibration.

No explicit obstacle integration. YOLOv8n detection exists but is not integrated into the planner cost function.

Conservative turn behavior. The containment guard prefers hold over unsafe turns; stale intent can cause unnecessary stops. Validation concentrated on VID_017.

Internal validity. The same campus is used for training and evaluation, risking geographic overfitting. The 20% validation hold-out and train/test split (held-out IoU 0.974 > training IoU 0.963) partially address this. Statistical significance confirmed by Friedman test ($p < 10^{-61}$, $N = 71$ common frames) and pairwise Wilcoxon tests with Bonferroni correction ($N = 72$ –448 depending on planner path availability).

External validity. Results may not generalize to non-campus environments, different surfaces, weather, or camera configurations. The 10 Hz target is validated on a workstation CPU; Raspberry Pi latencies are expected to be 2 – $3\times$ higher.

Construct validity. Lateral center error and inside-GT ratio are proxies. Temporal stability analysis (4.5 deg/frame jitter) partially addresses steering quality concerns.

CHAPTER 6

CONCLUSION AND FUTURE WORK

This thesis developed and evaluated a monocular-camera sidewalk navigation pipeline through four design iterations, culminating in a dual-domain architecture: image-space midpoint planning for straight-ahead following (14.5 px error, 3.0 ms, $p < 10^{-61}$) and BEV template-approval with GPS-conditioned turn planning for junction maneuvers.

6.1 Summary of Contributions

Contribution 1: A ten-method, four-family planner comparison on 448 hand-annotated frames (Chapter 4). Image-space midpoint achieves the lowest error and fastest runtime; its intent-agnostic nature motivates the dual-domain architecture.

Contribution 2: A template-approval planner replacing skeleton-graph construction with arc verification (Chapter 3), reducing heading error by 40.6% with 100% success on 1,800 frames.

Contribution 3: A teacher–student segmentation recipe (OneFormer Swin-L $\rightarrow$ SegFormer-B0) achieving IoU 0.964 after fine-tuning on 228 human-corrected masks, with held-out IoU (0.974) exceeding training IoU (0.963).

Contribution 4: A multi-scale evaluation protocol—448 annotated frames, intent evaluation on 17 930 frames, temporal stability analysis, and 1,800-frame sustained validation—with train/test split and Friedman/Wilcoxon statistical tests.

Contribution 5: A GPS-conditioned waypoint-turn planner with containment gating achieving 0% failure rate (Table 4.12).

6.2 Future Work

The following directions are organized by proximity to the current system state.

6.2.1 Near-Term Extensions

Closed-loop deployment. The most critical next step is deploying the complete stack on the physical scooter with real-time obstacle detection, measuring actual tracking error against ground-truth trajectories, and characterizing failure modes under closed-loop vehicle dynamics.

Multi-annotator evaluation. The current 448-frame evaluation set was produced by a single annotator. Adding additional annotators for inter-rater agreement analysis, and expanding to include frames from different seasons and times of day, would provide stronger evidence and address the single-annotator limitation noted in Section 5.4.

Cross-environment generalization. Evaluation on diverse sidewalk environments—different cities, weather conditions, surface materials, and nighttime illumination—would strengthen the external validity of the findings reported here.

6.2.2 Longer-Term Directions

Temporal video segmentation. Video-level consistency mechanisms such as temporal attention or optical-flow guidance could further reduce frame-to-frame segmentation flicker beyond what EMA smoothing achieves.

Multi-camera or depth fusion. Adding a depth sensor or stereo pair would enable metric obstacle distance estimation without the pinhole approximation, while retaining the monocular pipeline as the primary perception path.

Global route planning. The waypoint-turn planner introduced in this work provides GPS-conditioned junction behavior, but the GPS integration remains coarse (left/right/straight intent derived from bearing comparison). A full global route planner with a topological map, intersection detection, and multi-waypoint sequencing would enable autonomous multi-block navigation.

The geometric coverage insight at the core of this work—that monocular perspective images preserve denser planning information than BEV projections—applies to corridor-following tasks beyond sidewalk navigation. The modular, interpretable architecture presented in this thesis enables incremental improvement, as each component (segmentation, planning, temporal smoothing, turn handling) can be independently upgraded or replaced without disrupting the overall pipeline.

CHAPTER A

ALGORITHM PSEUDOCODE

This appendix provides formal pseudocode for the three planning algorithms developed in this thesis (Chapter 3). These methods combine established techniques—per-row boundary averaging, pre-computed trajectory libraries, and circular arc fitting—in a configuration tailored to monocular sidewalk navigation.

Algorithm 1 Image-Space Midpoint Planner

Require: Camera-space binary mask M_{img} ($H \times W$), ego column x_{ego}

Ensure: Smoothed centerline path $\mathbf{p}$

- 1: $M_c \leftarrow$ largest connected component of M_{img} touching ego band
 - 2: $\mathbf{p} \leftarrow \emptyset$
 - 3: **for** each row y from bottom to top **do**
 - 4: $\mathbf{x}_s \leftarrow$ nonzero pixel columns in $M_c[y]$
 - 5: **if** $|\mathbf{x}_s| < w_{\text{min}}$ **then continue**
 - 6: **end if**
 - 7: Group $\mathbf{x}_s$ into contiguous runs
 - 8: Select best run by width and proximity to reference center
 - 9: $x_L, x_R \leftarrow$ left and right boundaries of selected run
 - 10: $x_{\text{mid}} \leftarrow (x_L + x_R)/2$
 - 11: Append (x_{mid}, y) to $\mathbf{p}$
 - 12: **end for**
 - 13: $\mathbf{p} \leftarrow \text{SAVITZKYGOLAY}(\mathbf{p}, \text{window} = 9, \text{order} = 2)$
 - 14: **return** $\mathbf{p}$
-

Algorithm 2 Template-Approval Planner

Require: BEV mask M , previous family f_{prev} , GPS intent ι

Ensure: Approved path, confidence, family ID

Phase A: Corridor Extraction

- 1: **for** each sampled row r from ego upward, step 4 px **do**
- 2: Scan $M[r]$ for contiguous runs; select best by width and center proximity
- 3: Record boundaries (ℓ_r, r_r, c_r) ; update reference center
- 4: **end for**
- 5: Compute corridor confidence γ from coverage, span, width consistency

Phase B: Template Bank

- 6: $\mathcal{T} \leftarrow \{\text{straight, left-gentle, left-medium, left-sharp, right-gentle, right-medium, right-sharp}\}$
- 7: **for** each template **do** build circular arc from ego; resample at $\Delta s = 0.25$ m
- 8: **end for**
- 9: **if** $\iota \in \{\text{left, right}\}$ **then** filter $\mathcal{T}$ to matching family
- 10: **end if**

Phase C: Score Templates

- 11: **for** each $T_k \in \mathcal{T}$ **do**
- 12: $\rho_{\text{contain}} \leftarrow$ fraction of samples inside corridor
- 13: $\rho_{\text{near}} \leftarrow$ containment in first 1.2 m
- 14: $S_k \leftarrow 0.22\rho_{\text{contain}} + 0.18\rho_{\text{near}} + 0.16s_{\text{clear}} + 0.10s_{\text{center}} + 0.14s_{\text{cont}} + \dots$
- 15: $\text{approved}_k \leftarrow \rho_{\text{contain}} \geq 0.60 \wedge \rho_{\text{near}} \geq 0.72$
- 16: **end for**

Phase D: Hysteresis

- 17: Add family-reuse bonus +0.06 to templates matching f_{prev}
- 18: Apply straight-preference and switch-margin rules

Phase E: Decision

- 19: $T^* \leftarrow$ top-ranked approved candidate
- 20: **return** (path, approved, confidence, T^* , family)

Algorithm 3 GPS-Conditioned Waypoint-Turn Planner

Require: BEV corridor $\mathcal{C}$, GPS intent $\iota \in \{\text{left}, \text{right}\}$, BEV mask M

Ensure: Turn path, containment status, confidence

Step 1: Decision Band Scan

- 1: Scan forward band $[2.0, 7.0]$ m for asymmetric road openings on commanded side
- 2: $\sigma \leftarrow$ fraction of rows with commanded-side support exceeding opposite side

Step 2: Support Clustering

- 3: Group supported rows into clusters (max gap ≤ 0.5 m)
- 4: $\tau \leftarrow$ midpoint of largest cluster (candidate turn apex)

Step 3: Target Gating

- 5: **if** acquiring **then** require $\sigma \geq 0.40$ for 3 consecutive frames
- 6: **else if** sustaining **then** require $\sigma \geq 0.25$; release after 5 frames below
- 7: **end if**

Step 4: Circular Arc Path

- 8: Fit constant-curvature arc from ego through apex τ
- 9: $R \leftarrow \Delta x / \sin \theta$; parameterize as $x = R \sin \phi$, $y = \pm R(1 - \cos \phi)$

Step 5: Containment Gating

- 10: Require $\geq 60\%$ of path inside corridor AND $\geq 70\%$ in near-field (≤ 1.2 m)
 - 11: Exclude first 4 samples (near-ego BEV noise zone)
 - 12: **if** containment fails **then return** hold behavior with speed reduction
 - 13: **end if**
 - 14: **return** (arc path, confidence, slowdown recommendation)
-

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